

SAVITRIBAI PHULE PUNE UNIVERSITY
M.Sc. (Computer Application) Sem-II
Practical Examination 2023 - 2024
SUBJECT: CA 555 MJP: Lab course Based on CA 552 MJ
Introduction to Data Science

[Time: 3 Hours]

[Max. Marks: 35]

1. Write a R program to take input from the user (name and age) and display the values. Also print the version of R installation. [10 M]
2. Using R import the data from Excel/.CSV file and find mean, median, mode, quartiles. [20 M]
3. Viva. [5 M]

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1. Write a R program to create a simple bar plot of five subjects marks. [10 M]
2. Using R import the data from Excel/.CSV file and find standard deviation, variance and co-variance. [20 M]
3. Viva. [5 M]

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1. Write a R program to create a list containing a vector, a matrix and a list and remove the second element. [10 M]
2. Write a R program to call the (built-in) dataset air quality. Remove the variables 'Solar.R' and 'Wind' and display the data frame. [20 M]
3. Viva. [5 M]

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1. Write a R program to create an empty data frame. [10 M]
2. Write an R program to create a Data frames which contain details of 5employees and display the details in ascending order. [20 M]
3. Viva. [5 M]

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1. Write a R program to merge two given lists into one list. [10 M]
2. Consider Weather dataset i) Selecting using the column number ii)
Selecting using the column name iii) Make a scatter plot to compare Wind
speed and temperature. [20 M]
3. Viva. [5 M]

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1. Write a R program to get the unique elements of a given string and unique numbers of vector. [10 M]
2. Write a R program to compare two data frames to find the row(s) in first data frame that are not present in second data frame. [20 M]
3. Viva. [5 M]

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1. Write R program to find whether given number is positive or negative.
[10 M]
2. Write R program to read number and print corresponding day name in a week.
[20 M]
3. Viva.
[5 M]

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1. Create a Matrix using R and Perform the operations addition, subtraction, multiplication. [10 M]
2. Write a R program to create an ordered factor from data consisting of the names of months. [20 M]
3. Viva. [5 M]

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1. Write a R program to find nth highest value in a given vector. [10 M]
2. Write a script in R to create a list of students and perform the following
Give names to the students in the list.
Add a student at the end of the list.
Remove the first Student.
Update the second last student. [20 M]
3. Viva. [5 M]

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1. Write an R program to sort a Vector in ascending and descending order.
[10 M]
2. Write an R Program to calculate Decimal into binary of a given number.
Consider the inbuilt iris dataset

Create a variable “y” and attach to it the output attribute of the
“iris”dataset .
Create a barplot to breakdown your output attribute.
Create a density plot matrix for each attribute by class value. [20 M]
3. Viva. [5 M]

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1. Write an R program to extract first 10 English letter in lower case and last 10 letters in upper case and extract letters between 22nd to 24th letters in upper case. [10 M]
2. Write a script in R to create a list of students and perform the following
Give names to the students in the list.
Add a student at the end of the list.
Remove the first Student.
Update the second last student. [20 M]
3. Viva. [5 M]

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1. Write an R program to convert a given matrix to a list and print list in ascending order. [10 M]
2. Consider Weather dataset i) Selecting using the column number ii) Selecting using the column name iii) Make a scatter plot to compare Wind speed and temperature. [20 M]
3. Viva. [5 M]

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1. Write a R program to create a data frame using two given vectors and display the duplicated elements and unique rows of the said data frame.
[10 M]
2. Write an R Program to calculate Decimal into binary of a given number. Consider the inbuilt iris dataset i) Create a variable “y” and attach to it the output attribute of the “iris”dataset .ii) Create a barplot to breakdown your output attribute. iii) Create a density plot matrix for each attribute by class value.
[20 M]
3. Viva.
[5 M]

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1. Write a R program to multiply two vectors of integers type and length 3.
[10 M]
2. Write a R program to save the information of a data frame in a file and display the information of the file.
[20 M]
3. Viva.
[5 M]

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1. Write a R program to list containing a vector, a matrix and a list and give names to the elements in the list. [10 M]
2. Write a R program to create a factor corresponding to height of women data set, which contains height and weights for a sample of women. [20 M]
3. Viva. [5 M]